

All-Genus 2D $\mathcal{N} = 1$ Superconformal Blocks

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We develop an algorithm for numerically computing two-dimensional $\mathcal{N} = 1$ superconformal blocks at arbitrary genera, in arbitrary plumbing channels, and for arbitrary spin structures. The algorithm utilizes a Virasoro $\oplus$ Virasoro subalgebra of the tensor product of the $\mathcal{N} = 1$ superconformal algebra with an auxiliary free fermion, which allows us to reduce the superconformal blocks to Virasoro conformal blocks.

I. INTRODUCTION

Superconformal blocks are the central objects for computing correlators of superconformal field theories (SCFTs). Given the superconformal block on a spin Riemann surface, together with the conformal data of the SCFT, namely the conformal weights and the structure constants, correlators of a 2-dimensional $\mathcal{N} = 1$ SCFT on that spin Riemann surface can be computed through the superconformal block decomposition.

On spin Riemann surfaces of genus $g = 0$ or $g = 1$, algorithms for numerically computing the superconformal blocks are well developed. Analogously to the Zamolodchikov recursion relation for the sphere 4-point Virasoro conformal block [1, 2], the 4-point superconformal block on a sphere can be computed recursively using recursion relations in the central charge c or in the intermediate conformal weights h . These recursion relations have been worked out for both the case with all-NS external states [3–5] and that with Ramond external states [6, 7]. This c -recursion relation can be generalized to arbitrary plumbing channels and arbitrary genus as long as all intermediate states are in the NS sector [8, 9]. Meanwhile, for the blocks that involve Ramond intermediate states, the h -recursion relations can only be generalized to genus 1 [10] while the c -recursion relation does not exist beyond tree level, due to the fact that the large- h or large- c limit of the Ramond conformal blocks does not exist at higher genus. Algorithms for numerically computing these higher-genus Ramond blocks are absent in the literature.

In this letter, we present the first numerical algorithm for these higher-genus Ramond blocks. The algorithm is based on the fact that the $\mathcal{N} = 1$ superconformal algebra SCA, direct summed with a free Majorana fermion algebra F, admits two copies of commuting Virasoro subalgebras [11, 12]. These subalgebras were first discovered in the context of the AGT relation [13, 14] and have been used in the context of the super-Liouville/double Liou-

ville correspondence [15, 16]. Using this Vir $\oplus$ Vir subalgebra, the superconformal block can be reduced to a sum of products of two Virasoro conformal blocks, which can be computed numerically as a series in the plumbing parameters using the c -recursion relation developed in [8].

The rest of this paper is organized as follows. In Section II, we define the superconformal blocks using the plumbing fixture. In Section III, we review the details of the Vir $\oplus$ Vir subalgebra and the branching rules of SCA $\oplus$ F supermultiplets. In Section IV, we derive the relation between superconformal blocks and Virasoro blocks for blocks without a certain type of obstruction which we call odd Ramond loops. Section V extends the construction to blocks with odd Ramond loops using an auxiliary fermion insertion. We present numerical tests for the algorithm in Section VI and discuss applications and future directions in Section VII.

II. THE SUPERCONFORMAL BLOCKS

An arbitrary punctured Riemann surface can be constructed by gluing 3-punctured Riemann spheres using the plumbing fixture, which takes two punctures with local coordinates z_1, z_2 and identifies a pair of annular regions using

$$z_1 z_2 = q, \quad |q| \leq 1, \quad (1)$$

where q is a complex plumbing parameter. A genus- g , n -punctured Riemann surface can be constructed by plumbing $(3g - 3 + n)$ pairs of punctures, and its $(3g - 3 + n)$ complex moduli are given by those $(3g - 3 + n)$ plumbing parameters. One can then describe a punctured Riemann surface using a trivalent graph Γ together with a q -parameter for each internal edge. The vertices of Γ correspond to the 3-punctured spheres, while the edges connect pairs of plumbed punctures.

To specify a spin structure on the Riemann surface, we assign a sign η_e for each internal edge e in Γ , and additionally label each edge of Γ by NS or R. At each vertex, only edges of types (NS, NS, NS) or (NS, R, R) can be connected together. We call this labeled graph $\hat{\Gamma}$ a plumbing channel. The data of the spin Riemann surface is specified by $(\hat{\Gamma}, \{q_e, \eta_e\})$. The sets of all NS

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internal edges, R internal edges, (NS, NS, NS) vertices and (NS, R, R) vertices in $\hat{\Gamma}$ are denoted by $E_{\text{NS}}, E_{\text{R}}, V_{\text{NS}}$ and V_{R} , respectively. We also include in $\hat{\Gamma}$ the order of the three half-edges that connect to each vertex, with the first half-edge corresponding to the operator inserted at $z = \infty$, the second corresponding to $z = 1$, and the third corresponding to $z = 0$. On a (NS, R, R) sphere, we always take the NS half-edge to be in the first slot.

Plumbing two punctures in an SCFT correlator amounts to summing over a complete basis of intermediate states. These states organize into supermultiplets. We denote an NS superconformal primary by ϕ . A Ramond supermultiplet has 2 ground states $w^\pm$ that are related by G_0 :

$$G_0 w^\pm = i\beta e^{\mp i\pi/4} w^\mp, \quad (2)$$

where β is defined by $\beta^2 = \frac{c}{24} - h$. We take w^+ to be Grassmann even, and hence w^- will be Grassmann odd. A basis of states in a supermultiplet is given by

$$\begin{aligned} \mathbf{L}_{-A}\phi &\equiv L_{-n_1} \cdots L_{-n_\ell} G_{-r_1} \cdots G_{-r_p} \phi, \\ \mathbb{L}_{-A}w^\pm &\equiv L_{-n_1} \cdots L_{-n_\ell} G_{-r_1} \cdots G_{-r_p} w^\pm, \end{aligned} \quad (3)$$

in the NS and R sectors, respectively. Here, $n_i \in \mathbb{Z}, r_j \in \mathbb{Z} + \nu$, and $n_1 \geq \dots \geq n_\ell > 0, r_1 > \dots > r_p > 0$, where $\nu = 0$ in the R sector and $\nu = \frac{1}{2}$ in the NS sector. We further define the total level of $\mathbf{L}_{-A}, \mathbb{L}_{-A}$ by $|A| \equiv \sum_{i=1}^\ell n_i + \sum_{j=1}^p r_j$, and their Grassmann parity by $(-1)^A \equiv (-1)^p$.

After replacing each pair of plumbed punctures by a pair of local operators, the SCFT correlator can be reduced to a product of three-point functions on the sphere. On an (NS, NS, NS) sphere, all these three-point functions of descendants can be reduced to either $\langle \phi_1 \phi_2 \phi_3 \rangle$ or $\langle \phi_1 (G_{-\frac{1}{2}} \phi_2) \phi_3 \rangle$ by superconformal Ward identities. We define the normalized three-point forms ρ_a by

$$\rho_0(\phi_1, \phi_2, \phi_3) = 1, \quad \rho_1(\phi_1, G_{-\frac{1}{2}} \phi_2, \phi_3) = 1, \quad (4)$$

where the operators ϕ_1, ϕ_2, ϕ_3 are inserted at $\infty, 1, 0$, respectively. The normalized three-point functions $\rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3)$ of descendants are defined by reducing them to $\rho_0(\phi_1, \phi_2, \phi_3), \rho_1(\phi_1, G_{-\frac{1}{2}} \phi_2, \phi_3)$ using superconformal Ward identities. On an (NS, R, R) sphere, there are 4 independent three-point functions, and we define

$$\begin{aligned} \rho_0^{(\eta)}(\phi_1, w_2^+, w_3^+) &= 1, & \rho_0^{(\eta)}(\phi_1, w_2^-, w_3^-) &= \eta, \\ \rho_1^{(\eta)}(\phi_1, w_2^+, w_3^-) &= 1, & \rho_1^{(\eta)}(\phi_1, w_2^-, w_3^+) &= -i\eta, \end{aligned} \quad (5)$$

for $\eta = \pm$. Again, the operators $\phi_1, w_2^\pm, w_3^\pm$ are inserted at $\infty, 1, 0$, respectively, and the normalized three-point functions of descendants are defined through superconformal Ward identities.

On a spin Riemann surface specified by the plumbing data $(\Gamma, \{q_e, \eta_e\})$, the superconformal block with internal

weights $\{h_e\}_{e \in E_{\text{NS}}}, \{\beta_e\}_{e \in E_{\text{R}}}$ is defined as

$$\begin{aligned} \mathbf{F}_{\hat{\Gamma}}^{\{f, \eta, a\}} &= \sum_{\{A, \alpha\}} (-1)^{K_{\Gamma}(\epsilon)} \prod_{e \in E_{\text{NS}} \cup E_{\text{R}}} q_e^{|A_{e_1}|} \eta_e^{\epsilon_e} \\ &\times \prod_{e \in E_{\text{NS}}} \mathbf{G}_{h_e}^{A_{e_1} A_{e_2}} \prod_{e \in E_{\text{R}}} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_2} \alpha_{e_2}} \\ &\times \prod_{v \in V_{\text{NS}}} \rho_{a_v}(\mathbf{L}_{-A_{v_1}} \phi_{v_1}, \mathbf{L}_{-A_{v_2}} \phi_{v_2}, \mathbf{L}_{-A_{v_3}} \phi_{v_3}) \\ &\times \prod_{v \in V_{\text{R}}} \rho_{f_v}^{(\eta_v)}(\mathbf{L}_{-A_{v_1}} \phi_{v_1}, \mathbb{L}_{-A_{v_2}} w_{v_2}^{\alpha_{v_2}}, \mathbb{L}_{-A_{v_3}} w_{v_3}^{\alpha_{v_3}}), \end{aligned} \quad (6)$$

where f_v, η_v label the 3-point functions on the (NS, R, R) spheres, and a_v labels those on the (NS, NS, NS) spheres. The two half-edges an edge e contains are labeled e_1, e_2 , and the three half-edges connected to a vertex v are labeled v_1, v_2, v_3 according to their aforementioned order [17]. The sum is over all displayed descendant labels A and ground state labels α on each half-edge, subject to the conditions

$$\begin{aligned} |A_{e_1}| &= |A_{e_2}|, \quad e \in E_{\text{NS}} \cup E_{\text{R}} \\ (-1)^{A_{v_1} + A_{v_2} + A_{v_3}} &= (-1)^{a_v}, \quad v \in V_{\text{NS}}, \\ (-1)^{A_{v_1} + A_{v_2} + A_{v_3} + \alpha_{v_2} + \alpha_{v_3}} &= (-1)^{f_v}, \quad v \in V_{\text{R}}. \end{aligned} \quad (7)$$

We define ϵ_{v_i} to be the total Grassmann parity of the descendant Φ_{v_i} , which stands for $\mathbf{L}_{-A_{v_i}} \phi_{v_i}$ in the NS sector or $\mathbb{L}_{-A_{v_i}} w_{v_i}^{\alpha_{v_i}}$ in the R sector. The sign $(-1)^{K_{\Gamma}(\epsilon)}$ is the Grassmann sign that arises from moving the operator order from the pairwise insertion order to the displayed order in the 3-point functions [18]. $\mathbf{G}, \mathbb{G}$ are inverse Gram matrices in the NS and R sectors, which are defined as the inverse of

$$\begin{aligned} (\mathbf{G}_h)_{AB} &= \langle \langle \mathbf{L}_{-A} \phi_h | \mathbf{L}_{-B} \phi_h \rangle \rangle, \\ (\mathbb{G}_\beta)_{A\alpha, B\alpha'} &= \langle \langle \mathbb{L}_{-A} w_\beta^\alpha | \mathbb{L}_{-B} w_\beta^{\alpha'} \rangle \rangle, \end{aligned} \quad (8)$$

where the double left bracket stands for BPZ conjugation. h, β label the conformal weight and G_0 eigenvalues of the primaries, which are normalized such that $\langle \langle \phi_h | \phi_h \rangle \rangle = 1, \langle \langle w_\beta^+ | w_\beta^+ \rangle \rangle = 1$.

III. THE DOUBLE VIRASORO SUBALGEBRA

The key fact we utilize is that the enlarged algebra $\text{SCA}_\nu \oplus \mathbf{F}_\nu$ constructed from direct summing the $\mathcal{N} = 1$ superconformal algebra SCA_ν with shift $\nu \in \{0, \frac{1}{2}\}$ with an auxiliary free fermion algebra $\mathbf{F}_\nu$

$$\{\psi_r, \psi_s\} = \delta_{r+s, 0}, \quad \{\psi_r, G_s\} = [\psi_r, L_n] = 0, \quad (9)$$

where $r, s \in \mathbb{Z} + \nu$, admits two commuting Virasoro subalgebras [11, 12, 16]. We introduce the following operators

$$\begin{aligned} L_n^{\text{F}} &= \frac{1}{2} \sum_{r \in \mathbb{Z} + \nu} r : \psi_{n-r} \psi_r : + \frac{1-2\nu}{16} \delta_{n, 0}, \\ U_n &= \sum_{r \in \mathbb{Z} + \nu} \psi_{n-r} G_r. \end{aligned} \quad (10)$$

The generators of two commuting Virasoro subalgebras are given by

$$\begin{aligned} L_n^{(1)} &= \frac{b^{-1}L_n - (b^{-1} + 2b)L_n^F + U_n}{b^{-1} - b}, \\ L_n^{(2)} &= \frac{bL_n - (b + 2b^{-1})L_n^F + U_n}{b - b^{-1}}, \end{aligned} \quad (11)$$

where b is defined by $c = \frac{3}{2} + 3(b + b^{-1})^2$, and we have assumed that $b \neq 0, 1$. These generators $L_n^{(1,2)}$ satisfy $L_n^{(1)} + L_n^{(2)} = L_n + L_n^F$, and the two central charges of the two Virasoro algebras are given by

$$c^{(1)} = 1 + \frac{3Q^2}{1 - b^2}, \quad c^{(2)} = 1 + \frac{3Q^2}{1 - b^{-2}}, \quad (12)$$

where the Liouville background charge Q is defined such that $Q = b + b^{-1}$.

Let $\mathcal{M}_{\text{NS}}(h), \mathcal{M}_{\text{R}}(\beta)$ denote the tensor products of the auxiliary fermion Fock space with the corresponding SCFT Verma module. We denote the two R-sector ground states for the free fermion by u^a with $a = 0, 1$, which satisfy

$$\psi_0 u^a = \frac{1}{\sqrt{2}} u^{1-a}. \quad (13)$$

The highest weight states in $\mathcal{M}_{\text{NS}}(h), \mathcal{M}_{\text{R}}(\beta)$ are respectively $\mathbf{1} \otimes \phi$ and $u^a \otimes w^\alpha$. For generic momentum P , these

representations have branching rules [11, 12]

$$\begin{aligned} \mathcal{M}_{\text{NS}}(h) &\cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} V_{c(1)}(h_n^{(1)}) \otimes V_{c(2)}(h_n^{(2)}), \\ \mathcal{M}_{\text{R}}(\beta) &\cong \bigoplus_{\alpha=0,1} \bigoplus_{n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}} V_{c(1)}(h_n^{(1)}) \otimes V_{c(2)}(h_n^{(2)}). \end{aligned} \quad (14)$$

Here $V_c(h)$ is a Virasoro Verma module with highest weight h and central charge c . The highest-weight vectors are denoted by v_n and v_n^α . Their conformal weights are

$$\begin{aligned} h_n^{(1)} &= \frac{Q^2/4 - (P + 2nb)^2}{2(1 - b^2)}, \\ h_n^{(2)} &= \frac{Q^2/4 - (P + 2nb^{-1})^2}{2(1 - b^{-2})}, \end{aligned} \quad (15)$$

where the Liouville momentum P is defined using $P^2 = \frac{1}{4}Q^2 - 2h$ in the NS sector and $P^2 = 2\beta^2$ in the R sector. The NS $\text{Vir} \oplus \text{Vir}$ primary v_n has Grassmann parity $2n \bmod 2$ and level $2n^2$, and the R primary v_n^α has Grassmann parity $\alpha = 0, 1$ and level $2n^2 - \frac{1}{8}$. These $\text{Vir} \oplus \text{Vir}$ primaries v_n, v_n^α can be realized by a free field representation, which we review in appendix A.

IV. REDUCTION OF SUPERCONFORMAL BLOCKS

To reduce the superconformal block (6) to Virasoro conformal blocks, we first consider an enlarged $\text{SCA} \oplus \text{F}$ block $\hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f, \eta, a\}}$, defined as

$$\begin{aligned} \hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f, \eta, a\}} &= \sum_{\{A, \alpha, \mathbf{A}, \mathbf{a}\}} (-1)^{K_{\Gamma}(\epsilon + \epsilon')} \prod_{e \in E_{\text{NS}} \cup E_{\text{R}}} q_e^{|A_{e_1}| + |A_{e_1}|} \eta_e^{\epsilon_e + \epsilon'_e} (-1)^{\epsilon'_e} \prod_{e \in E_{\text{NS}}} \mathbf{G}_{h_e}^{A_{e_1} A_{e_2}} \prod_{e \in E_{\text{R}}} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_2} \alpha_{e_2}} \\ &\times \prod_{v \in V_{\text{NS}}} \hat{\rho}_{av}(\Psi_{-A_{v_1}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-A_{v_2}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_2}} \phi_{v_2}, \Psi_{-A_{v_3}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_3}} \phi_{v_3}) \\ &\times \prod_{v \in V_{\text{R}}} \hat{\rho}_{fv}^{(\eta_v)}(\Psi_{-A_{v_1}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-A_{v_2}} u^{a_{v_2}} \otimes \mathbb{L}_{-A_{v_2}} w_{v_2}^{\alpha_{v_2}}, \Psi_{-A_{v_3}} u^{a_{v_3}} \otimes \mathbb{L}_{-A_{v_3}} w_{v_3}^{\alpha_{v_3}}) \end{aligned} \quad (16)$$

where Ψ_{-A} denotes a chain of fermion creation operators ψ_{-r} with $r > 0$, and $|A|$ denotes its total level. The sum is over all displayed descendant and ground state labels, subject to

$$\begin{aligned} A_{e_1} &= A_{e_2}, \quad e \in E_{\text{NS}} \cup E_{\text{R}} \\ a_{e_1} &= a_{e_2}, \quad e \in E_{\text{R}} \\ (-1)^{A_{v_1} + A_{v_2} + A_{v_3}} &= 1, \quad v \in V_{\text{NS}} \\ (-1)^{A_{v_1} + A_{v_2} + A_{v_3} + a_{v_2} + a_{v_3}} &= 1, \quad v \in V_{\text{R}}, \end{aligned} \quad (17)$$

in addition to (7). ϵ and ϵ' denote the Grassmann parity of the SCA and the fermion part of the corresponding

descendant, respectively. The enlarged 3-point functions $\hat{\rho}_a, \hat{\rho}_f^{(\eta)}$ are defined as the product of the normalized 3-point function $\rho_f^{(\eta)}$, the free fermion 3-point function ρ_F , and the Grassmann sign $(-1)^{\epsilon_{v_2} \epsilon'_{v_3}}$, where the fermion 3-point function is normalized by $\rho_F(\mathbf{1}, u^0, u^0) = 1$ and $\rho_F(\mathbf{1}, u^1, u^1) = i$.

This enlarged block can be computed in two ways. One can take the basis $\Psi_{-A} \mathbf{1} \otimes \mathbb{L}_{-A} \phi$ for states in $\mathcal{M}_{\text{NS}}(h)$ and $\Psi_{-A} u^a \otimes \mathbb{L}_{-A} w^\alpha$ for $\mathcal{M}_{\text{R}}(\beta)$. This allows us to express the enlarged block in terms of the superconformal block

(6) and the free fermion block. Namely,

$$\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}} = \mathbf{F}_{\hat{\Gamma}}^{\{f,\eta,a\}} \star \mathbf{F}_{\hat{\Gamma}}^{\mathbf{F}}, \quad (18)$$

where we view the blocks as formal series in q_e, η_e , and the convolution $\star$ acts on monomials as

$$\begin{aligned} & \left(\prod_e q_e^{r_e} \eta_e^{\epsilon_e} \right) \star \left(\prod_e q_e^{s_e} \eta_e^{\epsilon'_e} \right) \\ &= (-1)^{K(\epsilon+\epsilon')-K(\epsilon)-K(\epsilon')+\sum_v \epsilon_v, 2\epsilon'_{v,3}} \prod_e q_e^{r_e+s_e} \eta_e^{\epsilon_e+\epsilon'_e}, \end{aligned} \quad (19)$$

while the free fermion block $\mathbf{F}_{\mathbf{F}}$ is defined as

$$\begin{aligned} \mathbf{F}_{\hat{\Gamma}}^{\mathbf{F}} &= \sum_{\{\mathbf{A}, \mathbf{a}\}} (-1)^{K_{\Gamma}(\epsilon')} \prod_{e \in E_{\text{NS}} \cup E_{\text{R}}} q_e^{|\mathbf{A}_{e1}|} (-\eta_e)^{\epsilon'_e} \\ &\times \prod_{v \in V_{\text{NS}}} \rho_{\mathbf{F}}(\Psi_{-\mathbf{A}_{v1}} \mathbf{1}, \Psi_{-\mathbf{A}_{v2}} \mathbf{1}, \Psi_{-\mathbf{A}_{v3}} \mathbf{1}) \\ &\times \prod_{v \in V_{\text{R}}} \rho_{\mathbf{F}}(\Psi_{-\mathbf{A}_{v1}} \mathbf{1}, \Psi_{-\mathbf{A}_{v2}} u^{\mathbf{a}_{v2}}, \Psi_{-\mathbf{A}_{v3}} u^{\mathbf{a}_{v3}}). \end{aligned} \quad (20)$$

As is, the enlarged block $\widehat{\mathbf{F}}_f^{(\eta, \eta')}$ vanishes whenever the labels f_v, η_v in $\rho_f^{(\eta)}$ around a closed Ramond loop $C \subset \hat{\Gamma}$ have product

$$(-1)^{\sum_{v \in C} f_v} \prod_{v \in C} \eta_v = -1. \quad (21)$$

We call such a loop an odd Ramond loop, and the algorithm for computing superconformal blocks with odd Ramond loops will be presented in Section V. In the rest of this section, we assume that there are no such odd Ramond loops.

The enlarged block can also be computed using the basis $L_{-A}^{(1)} L_{-B}^{(2)} v_n$ for $\mathcal{M}_{\text{NS}}(h)$ and $L_{-A}^{(1)} L_{-B}^{(2)} v_n^{\alpha}$ for $\mathcal{M}_{\text{R}}(\beta)$. This leads to the following decomposition of $\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}$:

$$\begin{aligned} \widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}} &= \sum_{\{n,\alpha\}} (-1)^{K(\epsilon)} \prod_{e \in E_{\text{NS}}} q_e^{2n_e^2} \eta_e^{2n_e} \prod_{e \in E_{\text{R}}} q_e^{2n_e^2 - \frac{1}{8}} \eta_e^{\alpha_e} \\ &\times \prod_{i=1}^2 F_{\Gamma}(h_{e_i}^{(i)}, c^{(i)}) \prod_{v \in V_{\text{NS}}} \mathbf{B}_{a_v}(n_{v1}, n_{v2}, n_{v3}) \\ &\times \prod_{v \in V_{\text{R}}} \mathbb{B}_{f_v}^{(\eta_v)}(n_{v1}, n_{v2}, n_{v3}, \alpha_{v2}, \alpha_{v3}), \end{aligned} \quad (22)$$

where the labels n take values in $n \in \frac{1}{2}\mathbb{Z}$ for NS half-edges, and $n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$ for R half-edges. $F(h_e; c)$ denotes the Virasoro conformal block in the same channel Γ with intermediate weights h_e and central charge c . ϵ here denotes the Grassmann parity of the $\text{Vir} \oplus \text{Vir}$ primary v_n or v_n^{α} . $\mathbf{B}_a, \mathbb{B}_f^{(\eta)}$ are the branching coefficients, defined as

$$\begin{aligned} \mathbf{B}_a(n_1, n_2, n_3) &= \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})}{\|v_{1,n_1}\| \|v_{2,n_2}\| \|v_{3,n_3}\|}, \\ \mathbb{B}_f^{(\eta)}(n_1, n_2, n_3, \alpha_2, \alpha_3) &= \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3})}{\|v_{1,n_1}\| \|v_{2,n_2}^{\alpha_2}\| \|v_{3,n_3}^{\alpha_3}\|}. \end{aligned} \quad (23)$$

The (NS, NS, NS) branching coefficients have been worked out in [16], while the (NS, R, R) coefficients have never been computed analytically. We now present a recursive algorithm for determining both numerically. It can be proved that, for $n > 0$, $L_1 v_n, L_1 v_n^{\alpha}$ both only contain level- $(4n-3)$ states in the $\text{Vir} \oplus \text{Vir}$ module generated by $v_{n-1}, v_{n-1}^{\alpha}$, while $L_{-1} v_n, L_{-1} v_n^{\alpha}$ only contain level-1 states in the v_n, v_n^{α} module as well as level- $(4n-1)$ states in the $v_{n-1}, v_{n-1}^{\alpha}$ module. Namely, there exist coefficients $\mathbf{V}_n^{(1,2)}, \mathbf{V}_{\pm 1,n}^{AB}, \mathbb{V}_n^{(1,2)}, \mathbb{V}_{\pm 1,n}^{AB}$ such that

$$\begin{aligned} L_1 v_n^{\alpha} &= \sum_{|A|+|B|=4n-3} \mathbb{V}_{1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^{\alpha}, \\ L_{-1} v_n^{\alpha} &= \mathbb{V}_{n,\alpha}^{(1)} L_{-1}^{(1)} v_n^{\alpha} + \mathbb{V}_{n,\alpha}^{(2)} L_{-1}^{(2)} v_n^{\alpha} \\ &\quad + \sum_{|A|+|B|=4n-1} \mathbb{V}_{-1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^{\alpha}, \\ L_1 v_n &= \sum_{|A|+|B|=4n-3} \mathbf{V}_{1,n}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}, \\ L_{-1} v_n &= \mathbf{V}_n^{(1)} L_{-1}^{(1)} v_n + \mathbf{V}_n^{(2)} L_{-1}^{(2)} v_n \\ &\quad + \sum_{|A|+|B|=4n-1} \mathbf{V}_{-1,n}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}. \end{aligned} \quad (24)$$

These coefficients $\mathbf{V}_n^{(1,2)}, \mathbf{V}_{\pm 1,n}^{AB}, \mathbb{V}_n^{(1,2)}, \mathbb{V}_{\pm 1,n}^{AB}$ can be determined directly by applying the free-field realization of $L_{\pm 1}$ on (A3), (A5). A similar relation exists for $n < 0$, where $L_{\pm 1} v_n, L_{\pm 1} v_n^{\alpha}$ include states in the $v_{n+1}, v_{n+1}^{\alpha}$ module.

After determining the coefficients $\mathbb{V}$ and $\mathbf{V}$, one can use the L_1 conformal Ward identities

$$\begin{aligned} & \hat{\rho}_f^{(\eta)}(L_1 v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) = \\ & \hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_{-1} v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) + \hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, L_{-1} v_{3,n_3}^{\alpha_3}), \\ & \hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_1 v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) = \\ & \hat{\rho}_f^{(\eta)}(L_{-1} v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) - \hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_{-1} v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \\ & \quad - 2\hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_0 v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) - \hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, L_1 v_{3,n_3}^{\alpha_3}), \end{aligned} \quad (25)$$

which become recursion relations for the branching coefficients $\mathbb{B}(n_1, n_2, n_3)$. Given the boundary values $\mathbb{B}(n_1, n_2, n_3)$ with $-\frac{3}{4} \leq n_i \leq \frac{3}{4}$, which can be computed analytically using the free field realization, one can solve for $\mathbb{B}(n_1, n_2, n_3)$ recursively.

Once one knows the branching coefficients, one can use the c -recursion relation to compute the Virasoro conformal blocks [8], which computes $\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}$ as a q -series. The free fermion block $\mathbf{F}_{\hat{\Gamma}}^{\mathbf{F}}$ can be computed from its definition efficiently due to the small size of the space of fermion descendants. The superconformal block $\mathbf{F}_{\hat{\Gamma}}^{\{f,\eta,a\}}$, as a series in $\{\eta_e, q_e\}$, can be extracted coefficient-by-coefficient from (18).

V. BLOCKS WITH ODD RAMOND LOOPS

We now turn to the case where the block contains odd Ramond loops. In this case, we define a new, non-vanishing enlarged block $\hat{\mathbf{F}}_f^{(f,\eta,a)}[\Theta\psi]$ by the following: For each odd Ramond loop C , we split one of the edges e in C . Suppose e connects two Riemann spheres with complex coordinates z, z' , and the punctures e connects to are located at $z = z_0, z' = z'_0$. We replace e by introducing a new Riemann sphere w , and glue the two punctures $w = 0, w = \infty$ with the two original spheres z_1, z_2 by

$$(z - z_0)w = q_{e_L}, \quad (z' - z'_0)w^{-1} = q_{e_R}. \quad (26)$$

The two new edges e_L, e_R formed are both Ramond. We repeat this splitting procedure for each odd Ramond loop in $\hat{\Gamma}$, and we denote the set of all split edges by E_{split} . This procedure defines a new plumbing channel $\hat{\Gamma}_{\text{split}}$, whose plumbing parameters are given by $\{q_e\}_{e \notin E_{\text{split}}}$ and $\{q_{e_L}, q_{e_R}\}_{e \in E_{\text{split}}}$. The original parameter q_e for a split edge can be related to q_{e_L}, q_{e_R} by $q_e = q_{e_L} q_{e_R}$. Similarly, the spin sign η_e can be recovered by $\eta_e = \eta_{e_L} \eta_{e_R}$.

On each new sphere w , we insert the operator $\Theta\psi$ at $w = 1$, where the operator Θ acts on the fermion ground states as

$$\Theta u^b = (-1)^{b+1} u^{1-b}, \quad (27)$$

and it anticommutes with ψ_r, G_r while commuting with L_n . The new enlarged block $\hat{\mathbf{F}}_f^{(f,\eta,a)}[\Theta\psi]$ is defined to be

$$\begin{aligned} \hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}[\Theta\psi] = & \sum_{\{A,\alpha,A,a\}} (-1)^{K_{\Gamma}(\epsilon+\epsilon')} \prod_{e \in E_{\text{NS}} \cup (E_{\text{R}} \setminus E_{\text{split}})} q_e^{|A_{e_1}| + |A_{e_1}|} \eta_e^{\epsilon_e + \epsilon'_e} (-1)^{\epsilon'_e} \prod_{e \in E_{\text{NS}}} \mathbf{G}_{h_e}^{A_{e_1} A_{e_2}} \prod_{e \in E_{\text{R}} \setminus E_{\text{split}}} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_2} \alpha_{e_2}} \\ & \times \prod_{v \in V_{\text{NS}}} \hat{\rho}_{a_v}(\Psi_{-A_{v_1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-A_{v_2}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_2}} \phi_{v_2}, \Psi_{-A_{v_3}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_3}} \phi_{v_3}) \\ & \times \prod_{v \in V_{\text{R}}} \hat{\rho}_{f_v}^{(\eta_v)}(\Psi_{-A_{v_1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-A_{v_2}} u^{a_{v_2}} \otimes \mathbb{L}_{-A_{v_2}} w_{v_2}^{\alpha_{v_2}}, \Psi_{-A_{v_3}} u^{a_{v_3}} \otimes \mathbb{L}_{-A_{v_3}} w_{v_3}^{\alpha_{v_3}}) \\ & \times \prod_{e \in E_{\text{split}}} q_{e_L}^{|A_{e_1}| + |A_{e_1}|} q_{e_R}^{|A_{e_2}| + |A_{e_2}|} \eta_e^{\epsilon_e + \epsilon'_e} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_L} \alpha_{e_L}} \mathbb{G}_{\beta_e}^{A_{e_R} \alpha_{e_R}, A_{e_2} \alpha_{e_2}} \\ & \times \langle \langle \Psi_{-A_{e_2}} u^{a_{e_2}} \otimes \mathbb{L}_{-A_{e_R}} w_{e_R}^{\alpha_{e_R}} | \Theta\psi(1) | \Psi_{-A_{e_1}} u^{a_{e_1}} \otimes \mathbb{L}_{-A_{e_L}} w_{e_L}^{\alpha_{e_L}} \rangle \rangle, \end{aligned} \quad (28)$$

where the sum is over all displayed labels, subject to (7) and (17), except that the fermion labels A_{e_1}, A_{e_2} and a_{e_1}, a_{e_2} may differ due to the $\psi(1)$ insertion. Their combined parities are constrained to be the same, namely

$$(-1)^{A_{e_1} + a_{e_1}} = (-1)^{A_{e_2} + a_{e_2}}, \quad e \in E_{\text{split}}. \quad (29)$$

This enlarged block is non-vanishing, and can be expressed as a convolution of the ordinary superconformal block and the free fermion block $\mathbf{F}_{\hat{\Gamma}}^F[\Theta\psi]$ with the same insertion $\Theta\psi(1)$ on the inserted spheres of each split edge. Namely,

$$\hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}[\Theta\psi] = \mathbf{F}_{\hat{\Gamma}}^{\{f,\eta,a\}} \star \mathbf{F}_{\hat{\Gamma}}^F[\Theta\psi]. \quad (30)$$

On the other hand, since the operator ψ can be regarded as $v_{1/2}(P)$ with Liouville momentum $P = \frac{Q}{2}$, and the operator Θ commutes with the two copies of Virasoro algebras, one can decompose $\hat{\mathbf{F}}_f^{(\eta,\eta')}[\Theta\psi]$ into Virasoro

blocks as

$$\begin{aligned} \hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}[\Theta\psi] = & \sum_{\{n,\alpha\}} (-1)^{K(\epsilon)} \prod_{e \in E_{\text{NS}}} q_e^{2n_e^2} \eta_e^{2n_e} \prod_{e \in E_{\text{R}} \setminus E_{\text{split}}} q_e^{2n_e^2 - \frac{1}{8}} \eta_e^{\alpha_e} \\ & \times \prod_{e \in E_{\text{split}}} q_{e_L}^{2n_{e_L}^2 - \frac{1}{8}} q_{e_R}^{2n_{e_R}^2 - \frac{1}{8}} \eta_e^{\alpha_e} \mathbb{S}(n_{e_R}, n_{e_L}, \alpha_e) \\ & \times \prod_{i=1}^2 F_{\Gamma'}(h_e^{(i)}, h_{\psi}^{(1)}, c^{(i)}) \prod_{v \in V_{\text{NS}}} \mathbf{B}_{a_v}(n_{v_1}, n_{v_2}, n_{v_3}) \\ & \times \prod_{v \in V_{\text{R}}} \mathbb{B}_{f_v}^{(\eta_v)}(n_{v_1}, n_{v_2}, n_{v_3}, \alpha_{v_2}, \alpha_{v_3}), \end{aligned} \quad (31)$$

where $F_{\Gamma'}(h_e^{(i)}, h_{\psi}^{(1)}, c^{(i)})$ stands for Virasoro conformal blocks in the new channel Γ' with external conformal weights

$$h_{\psi}^{(1)} = -\frac{1+2b^2}{2(1-b^2)}, \quad h_{\psi}^{(2)} = \frac{b^2+2}{2(1-b^2)}. \quad (32)$$

on the punctures with $\Theta\psi$ insertions. The splitting coef-

ficient $\mathbb{S}(n_R, n_L, \alpha)$ is defined as

$$\mathbb{S}(n_R, n_L, \alpha) = \frac{\langle v_{n_R}^\alpha | \Theta \psi(1) | v_{n_L}^\alpha \rangle}{\|v_{n_L}^\alpha\| \|v_{n_R}^\alpha\|}. \quad (33)$$

After knowing the coefficients $\mathbb{V}_{n,\alpha}^{1,2}, \mathbb{V}_{\pm 1, n, \alpha}^{AB}$ in (24), the splitting coefficients can be solved for using the L_1 conformal Ward identity

$$\langle L_1 v_{n_R}^\alpha | \Theta \psi(1) | v_{n_L}^\alpha \rangle = \langle v_{n_R}^\alpha | \Theta \psi(1) | L_{-1} v_{n_L}^\alpha \rangle, \quad (34)$$

which becomes a recursion relation for the splitting coefficients after plugging in (24). After solving for the splitting coefficients and the branching coefficients, one can compute $\hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f, \eta, a\}}[\Theta \psi]$ using this decomposition, and the physical superconformal block $\mathbf{F}_{\hat{\Gamma}}^{\{f, \eta, a\}}$ can be extracted from (30).

VI. NUMERICAL TESTS

Using the algorithm presented in Sections IV and V, one can compute arbitrary superconformal blocks as a series in the plumbing parameters. The results of our algorithm pass many cross-checks: The low-level coefficients of $q_1^{n_1} q_2^{n_2} q_3^{n_3}$ in $\mathbf{F}_f^{(\eta, \eta')}$ agree with direct computations from the definition (6) up to level $n_1, n_2, n_3 \leq 5$. The superconformal block in the same genus-2 theta channel but with all NS internal edges agrees with the result from c -recursion [9] up to total level 10. On a laptop and using 40-digit working precision, computing the genus-2 theta block $\mathbf{F}_0^{(\eta, \eta')}$ up to level $n_1, n_2, n_3 \leq 10$ takes 7 minutes with $\eta\eta' = 1$, and 27 minutes with $\eta\eta' = -1$. A direct computation using the definition (6) is estimated to take more than 6 hours for the same block.

Moreover, there are plumbing data $\{q_e, \eta_e\}$ in the all-NS theta channel and the NS-R-R theta channel that describe the same spin Riemann surface, and we have checked that the ratio

$$\frac{Z_{\text{super-Liouville}}}{(Z_{\text{free boson+fermion}})^{1+2Q^2}} \quad (35)$$

of super-Liouville and free theory partition functions agrees in the two plumbing channels when evaluated using our superconformal blocks. The details of these cross-checks are included in the supplemental materials [19].

[Unfinished: Mercedes channel tests, defining the channels]

VII. DISCUSSION

In this work, we develop the algorithm for numerically computing superconformal blocks at arbitrary genus, in arbitrary plumbing channels, for arbitrary spin structures. The double Virasoro subalgebra allows us to decompose the enlarged $\text{SCA}_\nu \oplus \text{F}_\nu$ block into Virasoro conformal blocks, with the coefficients given by branching

coefficients, which are known on an (NS, NS, NS) sphere and can be solved numerically using a recursion relation on an (NS, R, R) sphere.

Our algorithm provides new tools for higher-loop superstring amplitudes. For 2-dimensional Type 0B and Type 0A string theory, such amplitudes can be used to further test the duality with matrix quantum mechanics [20–23], as well as the T-duality between the two theories [24]. They also permit direct worldsheet tests of matrix-integral predictions for recently proposed super Virasoro minimal strings [25–29]. One particularly interesting direction is the 2-dimensional $\text{SO}(23)$ heterotic string [30], where the matrix quantum mechanical dual remains unknown, and the higher-loop or higher-point amplitudes can be used to find it. In these applications of string amplitudes, our algorithm can be used to compute the integrand on moduli space. The integration can be performed using a similar method to [31]: For genus $g \leq 3$, one samples the moduli space parametrized by the period matrix and converts the period matrix into plumbing parameters using a similar algorithm to [32].

Beyond string amplitudes, modular covariance of higher-genus partition functions can be used to constrain the spectrum and the structure constants of $\mathcal{N} = 1$ SCFTs, extending the analysis of [33]. The superconformal blocks themselves are conjectured to be the wavefunctions of 3D supergravity in AdS_3 [29], and the numerical values of the blocks obtained from our algorithm can be used to compute gravitational path integrals in this theory [34].

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Appendix A: Free field representation of $\text{Vir} \oplus \text{Vir}$ primaries

The $\text{Vir} \oplus \text{Vir}$ primaries can be directly constructed using a free field realization of the $\text{SCA}_\nu \oplus \text{F}_\nu$ algebra [16]. Using free boson operators $\{c_n | n \neq 0\}$, free fermion operators $\{\eta_r | r \in \mathbb{Z} + \nu\}$ and the central element $\hat{P}$ that satisfy the algebraic relations

$$[c_n, c_m] = n\delta_{m+n,0}, \quad \{\eta_r, \eta_s\} = \delta_{r+s,0}, \quad (\text{A1})$$

the generators of the superconformal algebra SCA_ν can be realized as

$$\begin{aligned} L_{n \neq 0} &= \frac{1}{2} \sum_{k \neq 0, n} c_k c_{n-k} + \frac{1}{2} \sum_r r \eta_{n-r} \eta_r + \frac{i}{2} (Qn - 2\hat{P}) c_n, \\ L_0 &= \sum_{k > 0} c_{-k} c_k + \sum_{r > 0} r \eta_{-r} \eta_r + \frac{1}{2} \left(\frac{Q^2}{4} - \hat{P}^2 \right) + \frac{1-2\nu}{16}, \\ G_r &= \sum_{n \neq 0} c_n \eta_{r-n} + i(Qr - \hat{P}) \eta_r. \end{aligned} \quad (\text{A2})$$

Each free boson mode c_n and free fermion mode η can be expressed directly in terms of G_{-r} , L_{-n} and $\hat{P}$.

It can be proved that $\mathcal{M}_{\text{NS}}(h)$ is isomorphic to the Fock space generated by acting with $c_{n < 0}$, $\psi_{r < 0}$, $\eta_{r < 0}$ on a ground state $\phi(P)$ that satisfies $\hat{P}\phi(P) = P\phi(P)$ [35], and the $\text{Vir} \oplus \text{Vir}$ primaries in $\mathcal{M}_{\text{NS}}(h)$ can be constructed

using the free field operators $\chi_r(P) = \psi_r - i\eta_r(P)$ as

$$v_n(P) = \chi_{-\frac{4n-1}{2}}(P) \chi_{-\frac{4n-3}{2}}(P) \cdots \chi_{-\frac{1}{2}}(P) \phi(P) \quad (\text{A3})$$

where $\eta_r(P)$ is the solution of η_r in terms of $L_n, G_r, \hat{P}$ and with $\hat{P}$ replaced by P . v_n with negative n is defined by the reflection formula

$$v_n(P) = v_{-n}(-P). \quad (\text{A4})$$

Similarly in the R-sector, the $\text{Vir} \oplus \text{Vir}$ primaries in $\mathcal{M}_{\text{R}}(\beta)$ can be constructed by

$$\begin{aligned} v_n^{2n+\frac{1}{2} \bmod 2}(P) &= \chi_0(P) \chi_{-1}(P) \cdots \chi_{-(2n-\frac{1}{2})}(P) (u^0 \otimes w_\beta^+), \\ v_n^{2n-\frac{1}{2} \bmod 2}(P) &= \chi_0(P) \chi_{-1}(P) \cdots \chi_{-(2n-\frac{1}{2})}(P) \chi_0(-P) (u^0 \otimes w_\beta^+), \end{aligned} \quad (\text{A5})$$

for $n > 0$, and by the reflection formula

$$(v_n^\alpha(P))_+ = (v_{-n}^\alpha(-P))_+, \quad (v_n^\alpha(P))_- = -(v_{-n}^\alpha(-P))_-, \quad (\text{A6})$$

for $n < 0$. The subscripts $\pm$ denote the components ending in w_β^+ and w_β^- , respectively.

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